

MATTEO CAVAGNINO

CUDA Software Engineer | C++ | GPU Acceleration | SAR Simulation

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

CUDA Software Engineer with **3+ years of professional experience** building performance-critical C++/CUDA software for SAR simulation, 3D computer vision, and real-time systems. At MetaSensing, responsible for the architecture and development of a SAR simulator built largely from the ground up, with **5-10x performance gains** delivered on targeted GPU workloads.

EXPERIENCE

CUDA Software Engineer | MetaSensing

Milan, Italy | July 2025 - Present

- Own the architecture and development of a SAR simulator built largely from the ground up using **C++**, **CUDA**, and **NVIDIA OptiX**.
- Develop high-performance C++/CUDA software for SAR systems and other compute-intensive internal applications.
- Optimized targeted GPU workloads and CUDA components, delivering approximately **5-10x speedups**.
- Develop and tune software for modern NVIDIA GPUs and multi-GPU configurations, with emphasis on parallel efficiency and scalability.

3D Computer Vision Engineer | ACTEON GROUP / De Götzen

Fagnano Olona (VA), Italy | September 2023 - March 2025

- Developed performance-critical software for a real-time 3D scanning system using **C++**, **CUDA**, and **OpenGL**.
- Worked across point-cloud registration, 3D reconstruction, mesh alignment, and post-processing.
- Built components spanning GPU computing, computer vision, 3D geometry processing, and real-time rendering.

Software Support Engineering Specialist I | CAE

Sesto Calende (VA), Italy | December 2022 - September 2023

- Developed C++ and OpenGL software for civil and military full-flight helicopter simulators.
- Contributed to graphics-engine and flight-control software in complex real-time simulation systems.
- Worked in production codebases with strict reliability, maintainability, and real-time execution requirements.

TECHNICAL SKILLS

Core

C++, CUDA, NVIDIA OptiX, GPU Computing, HPC, Multi-GPU, OpenMP

Vision & Graphics

OpenCV, OpenGL, GLSL, Point Clouds, 3D Reconstruction, Mesh Processing

Libraries & Tooling

Eigen, GLM, Assimp, Asio, Boost, STL, CMake, Git, Visual Studio, VS Code

Additional

Python, C#, Java, JavaScript, SQL

Languages

English - C1 Certification

EDUCATION

M.Sc. in Computer Science

University of Milan
2019 - 2022 | 105/110

B.Sc. in Computer Science

University of Insubria
2016 - 2019 | 105/110

SELECTED PROJECTS

ShakEngine | GitHub

Solo C++ 2D game engine built from scratch with OpenGL, SFML, Dear ImGui, and GLSL; includes a custom ECS, collision, resource management, sprite batching, scene lifecycle, and audio.

Zombie++ | Portfolio

Solo Unity/C# 3D first-person survival game with modular, configurable systems for enemies, perks, weapons, and maps.